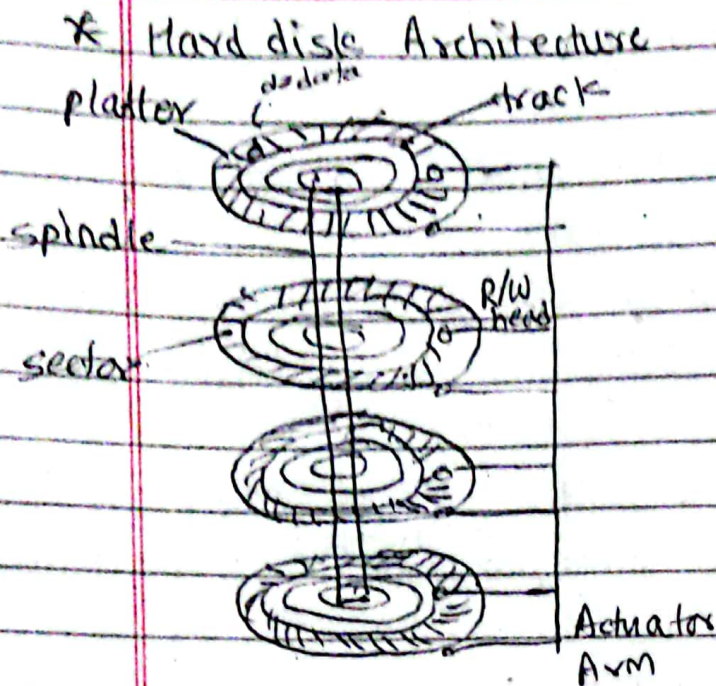


CHAPTER 6

I/O Device Management



- * Spindle is used to make platter move, either ^{or rotate} rightwise (clockwise) or anticlockwise. but will be unidirectional.
- * Above the platters, we have r/w head connected to an actuator arm which is used to fetch the data. Their movement is only backward & forward.
- * R/W head is holding both upper & lower surfaces of platter
- * There are tracks on the platter like CDs which have same numbers in both upper & lower surfaces
- * On every track, we have fixed no. of sectors
- * Inside these sectors, we put the data,
- * Platter → Surfaces → Track → Sectors → Data.

Qo. No. of platters = 8

Total surfaces = 8×2 (upper & lower)

No. of Tracks = ~~256~~ 256

∴ Total no. of tracks = $8 \times 2 \times 256$

1 track = 512 sectors

Total no. of sectors = $8 \times 2 \times 256 \times 512$

Data per sector = 512 kb

Total disk size = Platter $\times$ Surface $\times$ Track $\times$ Sector $\times$ Data

$$= 8 \times 2 \times 256 \times 512 \times 512 \text{ KB} \quad (1 \text{K} = 2^{10})$$

$$= 2^3 \times 2^1 \times 2^8 \times 2^9 \times 2^9 \times 2^{10} \text{ B}$$

$$= 2^{40} \text{ B}$$

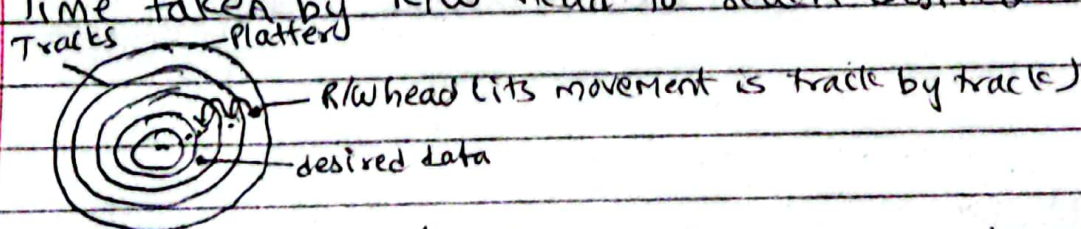
$$= \frac{1 \text{ TB}}{1024}$$

Here, no. of bits required to represent disk size = 40

* Disk Access Time

① Seek Time:

2) Time taken by R/W head to reach desired track.

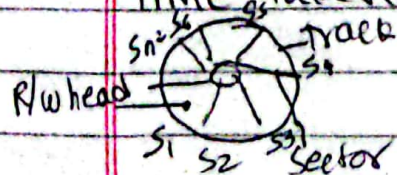


2) If whatever amt of seek time we can decrease, that much of performance will get improved

4) Firstly, you have to reach to a desired track, where the time taken to reach to desired track is seek time.

② Rotation Time:

2) Time taken for one full rotation (360°)



Seek time और track समान त पहुँच, और data को sector में है. rotation लगातार

③ Rotational Latency:

↳ Time taken to reach to desired sector. (half of rotation)

Best case $\rightarrow 0$ rotation

Worst case $\rightarrow$ full (1) rotation

Average case $\rightarrow$ half rotation

④ Transfer Time:

↳ Data to be Transferred

Transfer Rate

Transfer Rate : per second how much data i can transfer

Transfer Rate = No. of heads X Capacity of one Track X
(surfaces) (Sectors X data)

No. of Rotations in one second

$$\# \text{ Disk Access Time} = ST + RT + TT + CT + QT$$

/ \ \

* $\rightarrow$ Optional acc. to qn. Rotational latency/time Controller Time Queue Time

* Disk Scheduling Algorithm

↳ The goal is to minimize the seek times.

① FCFS (First Come First Serve)

② SSTF (Shortest Seek Time First)

③ SCAN

④ LOOK

CSCAN (Circular SCAN)

CLOOK (Circular LOOK)

Adv.
no starvation

Disadv.
performance

(without looking at data it goes to 50 value comes more compared to other)

① FCFS:

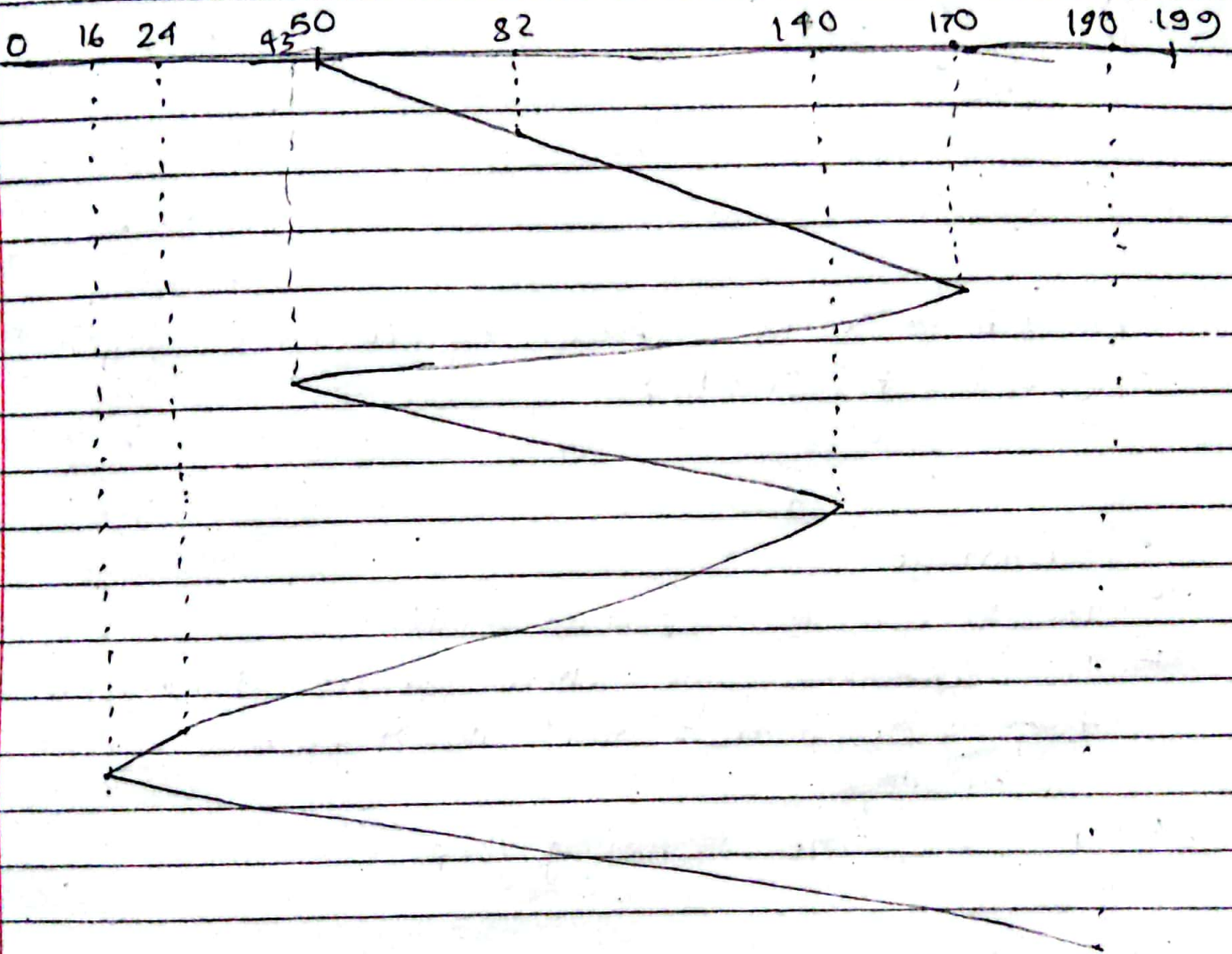
Qn. A disk contains 200 tracks (0-199).

Request queue contains track no.

82, 170, 43, 140, 29, 16, 190 respectively.

Current position of R/W head = 50

Calculate total no. of tracks movement by head.



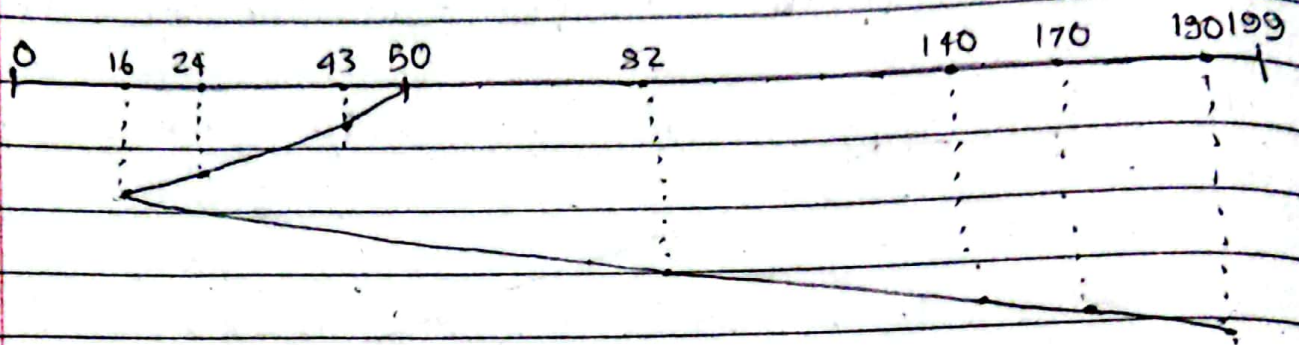
$$\begin{aligned} &= (82-50) + (170-82) + (170-43) + (140-43) + (140-29) + \\ &\quad (29-16) + (190-16) \quad \text{OR} \\ &= (170-50) + (170-43) + (140-43) + (140-16) + (190-16) \\ &= 642 \end{aligned}$$

② SSTF:

Qn: Same, Request: 82, 170, 43, 140, 24, 16, 190

Total no. of tracks movement by R/W head using SSTF?

If R/W head takes 1 ms to move from one track to another, then total time taken = ?



$$\begin{aligned} \therefore \text{Total no. of tracks movement by R/W head using SSTF} \\ &= (50 - 16) + (190 - 16) \\ &= 208 \end{aligned}$$

• Advantage

↳ tries to give the optimal result.

⇒ From wherever point where we are starting, we have to go to the nearest track no. & we have to move one by one

↳ Response time is coming good

• Problem

↳ Starvation (It maybe ~~one~~ ^{that} one request is waiting for a long time for its turn)

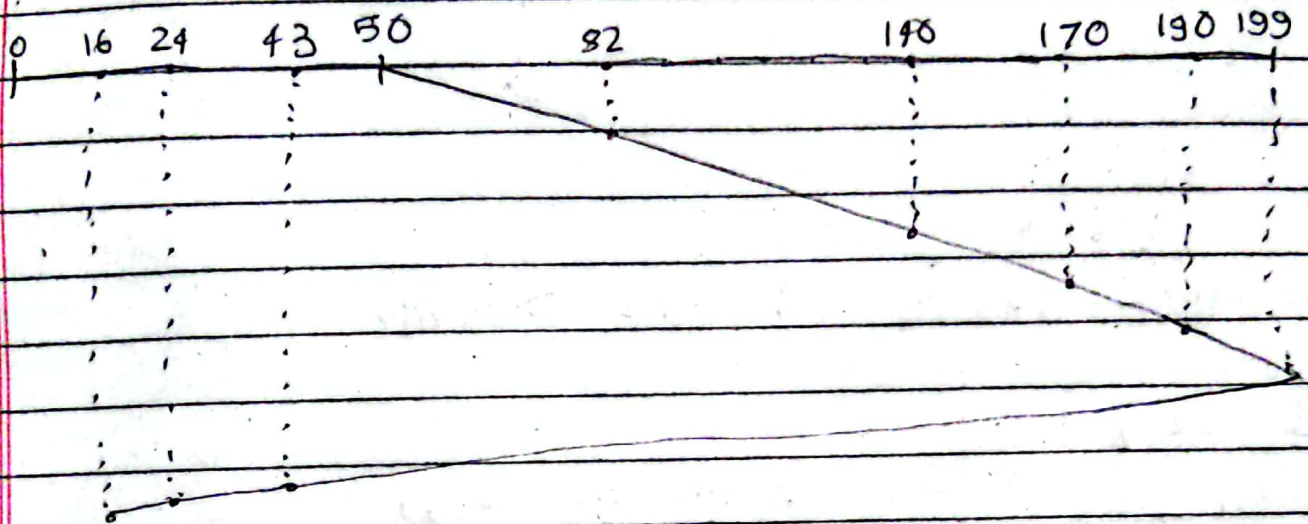
↳ Generation of overhead (we have to find nearest of n , nearest of $n-1$, $n-2$ & so on).

At every step you'll have to find who is nearest to that particular value

③ SCAN

Request: 82, 170, 43, 140, 24, 16, 190

2) SCAN alg. is also known as elevator algorithm where we have to move in one direction & move till last point in that direction



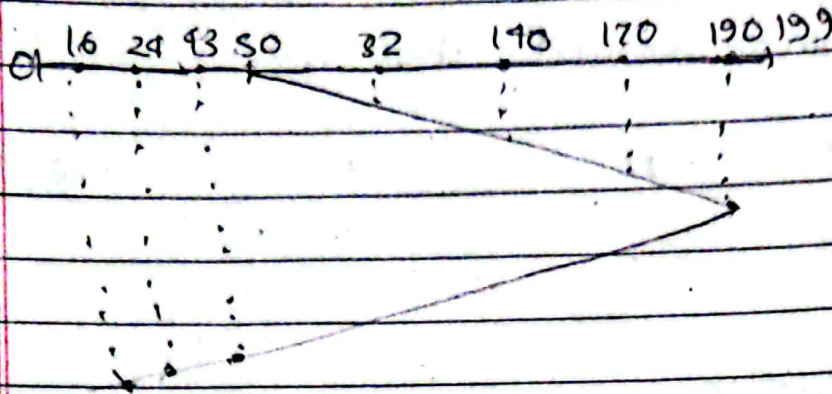
Total no. of tracks movement by R/W head using SCAN

$$= (199 - 50) + (199 - 16)$$
$$= 332$$

↳ It is going until 199 because maybe a request comes dynamically at run time.

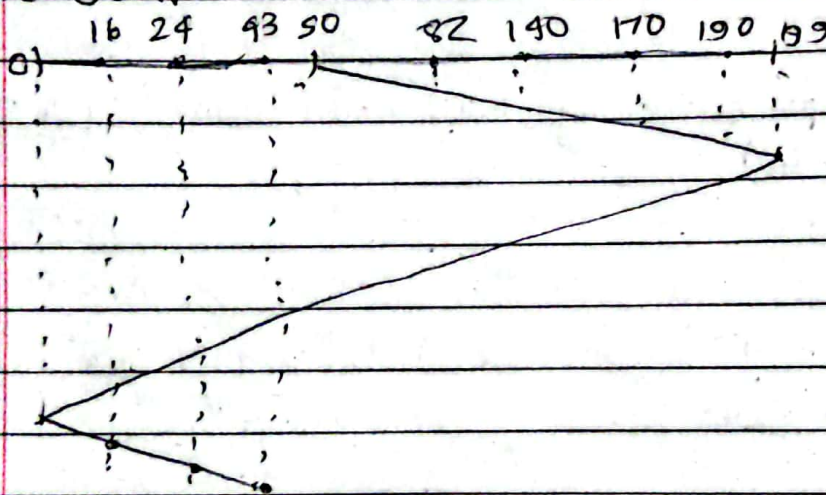
④ LOOK:

↳ last sama jadaina, 199 sama jadaina 190 batai farkinxa natra SCAN nai ho.



$$\Rightarrow (190 - 50) + (190 - 16) = 314 > \text{SCAN}$$

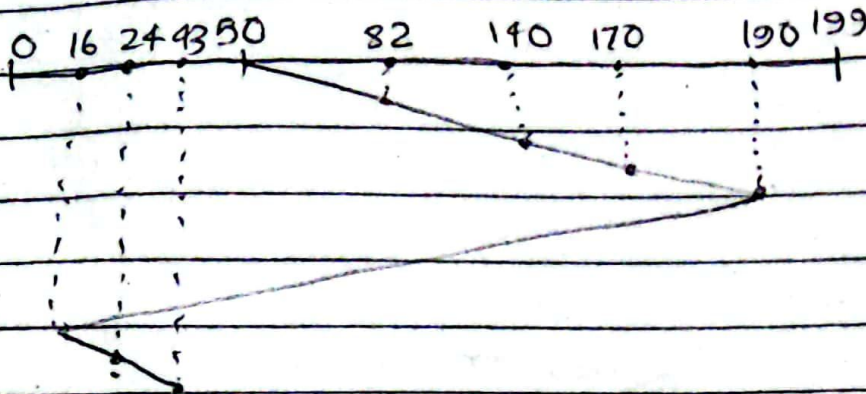
⑤ C-SCAN:



$$(199 - 50) + (199 - 0) + (43 - 0) = 391$$

↳ last sama janxa, feri 0 mai aunka mi R/W head ke current position vanda uta wa eta jun jun baloi se tei bata lanxa ~~wa~~ ulcerverse

⑥ C-LOOK



$$(190 - 50) + (190 - 16) + (43 - 16)$$

$$= 341$$

- 1) When we are changing the direction, we don't have to fulfill any request
- 2) last samna va fer^{last} O sama jadana, request mai depend hunxa